

AutoFOAM: The Self-Refining Autonomous OpenFOAM Agent

Bridging Natural Language and Computational Fluid Dynamics

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Outline

- 1 Motivation
- 2 System Architecture
- 3 Dataset Curation
- 4 Self-Evolution Pipeline
- 5 Results
- 6 Limitations & Future Work
- 7 Conclusion

The OpenFOAM Expertise Barrier

Why CFD is Hard to Access

- OpenFOAM: unmatched flexibility for physics simulation
- Requires mastery of dozens of **C++ dictionaries**
- Syntax is **strict and deeply interconnected**
- Moving from physics idea → working simulation is tedious and error-prone

Where Generic LLMs Fall Short

- Invent non-existent OpenFOAM keywords
- Inconsistent boundary conditions across files
- Lack physical understanding of flow regimes

The Vision

"Simulate a turbulent flow over a NACA0012 airfoil at $Re=10^6$ "

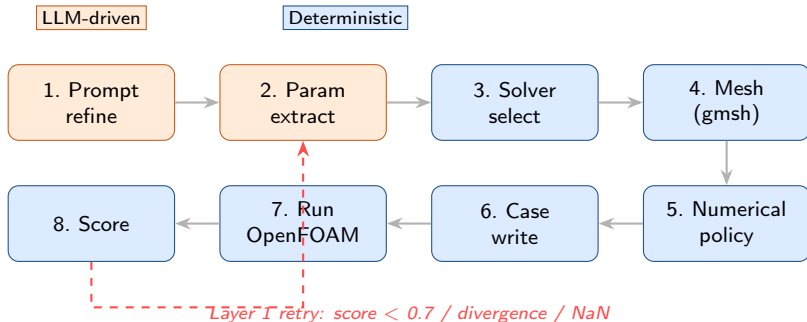


Full, convergent OpenFOAM case
generated autonomously

Related Work

FoamGPT, OpenFOAMGPT, Foam-Agent 2.0
— growing momentum for LLM-driven CFD

AutoFOAM: The Eight-Stage Pipeline



LLM is invoked only where needed

Prompt refinement, parameter extraction, RAG context, retry generation

Deterministic steps ensure correctness

Solver routing, mesh generation, case writing, execution, scoring

Key Design Choices

Constrained Parameter Extraction

- **vLLM** for high-throughput inference
- **xgrammar** JSON schema enforces `CFDParams` structure
- Output: geometry class, Re , fluid properties, solver settings, turbulence model
- Post-generation regex validates $Re = UL/\nu$ consistency

Deterministic Solver Routing

- Decouples solver choice from LLM
- Uses physics flags: transient, compressible, thermal, multiphase
- Routes to one of **7 approved solvers**
- Prevents LLM bias (e.g. choosing `simpleFoam` for transient vortex shedding)

Multi-Objective Reward $r \in [0, 1]$

Convergence (+0.40) | Residual $< 10^{-4}$ (+0.20) | Mass conservation (+0.05) | Correct solver (+0.10) | Valid BCs (+0.05)

Penalties: mesh quality, slow runtime, stagnation (−0.25 total)

Training Data: Static + Dynamic

Foundational Knowledge Corpus

- **402 rows** of expert OpenFOAM interaction trajectories
- 252 unique prompts covering all solver types and geometry parameterizations
- Accepted trajectories: score $r \geq 0.5$
- Format: natural language $\rightarrow$ summary $\rightarrow$ full C++ dictionary blocks

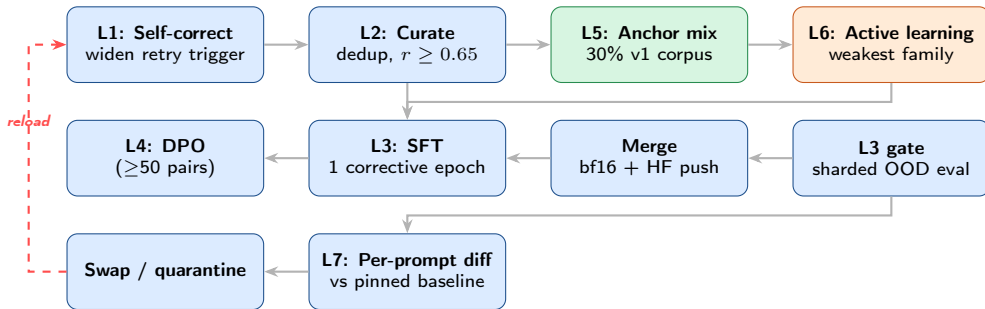
Dynamic Trajectory Capture

- Live production runs feed the dataset continuously
- High-success runs ($r \geq 0.65$): added to training pool
- Currently **210 de-duplicated topologies**
- Failed runs logged with failure type for DPO preference mining

Out-of-Distribution (OOD) Benchmark — 110 Prompts

Solver Family	Count	Share
simpleFoam (steady incompressible)	74	67%
rhoSimpleFoam (steady compressible)	9	8%
interFoam (multiphase VOF)	7	6%
Others (buoyantSimpleFoam, pimpleFoam, ...)	20	18%

The Seven-Layer Evolution Loop



L5: Anchor Mixing

30% foundational corpus injected into every retraining batch — prevents catastrophic forgetting

L6: Active Learning

Targets the weakest solver family; auto-generates new seeds to strengthen it

L7: Regression Firewall

Word-by-word diff against pinned baseline before any weight promotion

Anti-Collapse Mechanisms

The Self-Distillation Collapse Risk

Training on self-generated data can rapidly degrade model quality — AutoFOAM deploys **three complementary defences**.

RAG-Augmented Retry

Chroma vector DB of verified OpenFOAM tutorial configs.

Injects relevant dictionary snippets into the retry prompt.

Surgical Dictionary Patching

On localised FOAM FATAL errors, patches only the offending dictionary.

Skips costly gmsh remeshing — fast feedback loop.

Prompt Diversity Paraphrasing

Varies semantic structure of training prompts to prevent the model from overfitting to one phrasing pattern.

Layer 4: Direct Preference Optimization (DPO)

Failure trajectory (rejected) $\longleftrightarrow$ Layer-1-corrected success (chosen)
DPO explicitly penalises the syntactic formulations that caused divergence.

Autonomous Case Generation

Coverage

- **7 solver families:** simpleFoam, pimpleFoam, rhoSimpleFoam, rhoPimpleFoam, buoyantSimpleFoam, interFoam, icoFoam
- **13 parameterised gmsh geometries:** driven cavity, backward step, NACA airfoil, cylinder, periodic hill, S-bend, Ahmed body, ...
- Input: *one natural-language sentence*
- Output: correct polyMesh boundaries + convergent solution

OOD Headline Metrics

Metric	Value
FOAM FATAL-free	110/110
Execution success	100%
Solver routing accuracy	96.4%
Mean reward r	0.64
Geometry templates	13/13
Solver families	7/7

Misrouted Cases

Only 4/110 solver mismatches — all in **ambiguous low-Re regimes** (e.g. transient cylinder at $Re = 250$: icoFoam selected instead of pimpleFoam — physically justified).

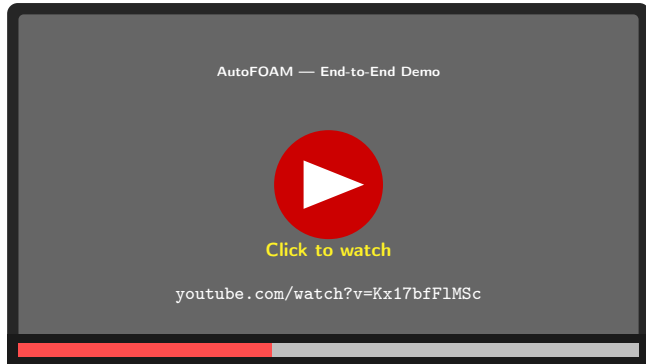
Representative OOD Prompts

AutoFOAM correctly maps natural language $\rightarrow$ physics $\rightarrow$ solver

Tag	Prompt	Predicted Solver	Match
cav_re150	<i>Cavity flow $Re=150$ in water, 0.4 m square</i>	simpleFoam	✓
pipe_lam	<i>Laminar pipe flow $Re=500$, diameter 0.02 m, water</i>	simpleFoam	✓
naca_re1m	<i>NACA0012 at $Re=10^6$, 5° angle of attack</i>	simpleFoam	✓
dam_tall	<i>Tall dam break, 2 m wide, 5 m water column collapse</i>	interFoam	✓
cdnoz_ma04	<i>Convergent-divergent nozzle with Mach 0.4 throat, air</i>	rhoSimpleFoam	✓
cyl_re250	<i>Transient cylinder $Re=250$ in water, $D=0.1$ m</i>	icoFoam	×

Evaluation Infrastructure

8-way sharded cluster of **NVIDIA H100** GPUs — 110 OOD prompts in isolation from training data



What the Demo Shows

- Natural-language prompt input
- Automated parameter extraction
- gmsh mesh generation
- OpenFOAM case file synthesis
- Live solver run + convergence
- Velocity contour output

Watch Online

youtu.be/Kx17bfFlMSc

Automatically Generated Meshes

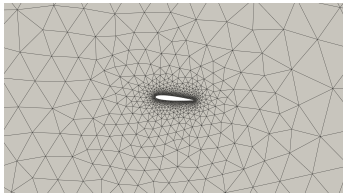


Figure: NACA-4-digit airfoil

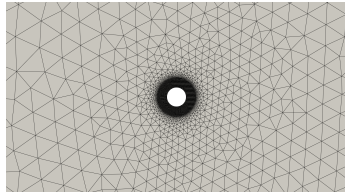


Figure: Cylinder in cross-flow

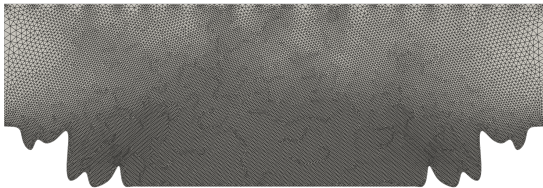


Figure: Periodic hill (Mellen polynomial)

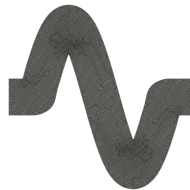


Figure: S-bend duct

Velocity Contours — Autonomous Flow Solutions

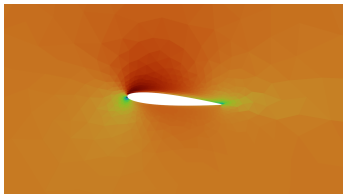


Figure: NACA-4-digit airfoil — velocity magnitude

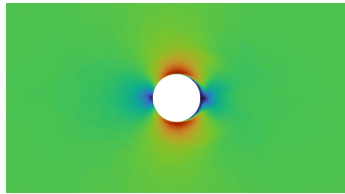


Figure: Cylinder in cross-flow — velocity magnitude

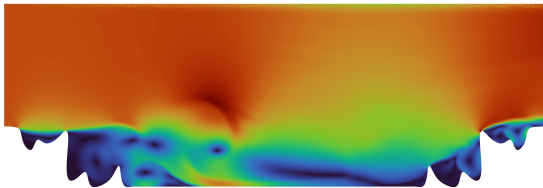


Figure: Periodic hill — velocity magnitude

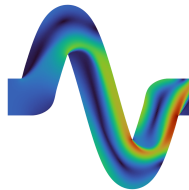


Figure: S-bend — velocity magnitude

Scope, Limitations, and Next Steps

Current Limitations

- **Geometry:** restricted to 13 parameterised gmsh models — no raw CAD (STEP/IGES) support
- **Physics:** RANS-only turbulence; no LES/DNS
- **Reward:** biased toward mathematical convergence — a numerically sound solution can still be *physically wrong* (e.g. incorrect drag coefficient)
- No cross-reference against benchmark databases (e.g. Ghia cavity data)

Future Work

- **Physics Validator:** external module comparing agent output against known benchmark data — closes the physics-validation gap
- CAD-to-mesh pipeline for complex industrial geometries
- LES/DNS numerical policy
- Broader solver coverage (e.g. `reactingFoam`, `chtMultiRegionFoam`)
- Tighter DPO negative feedback once physics validator is live

Conclusion

What We Built

AutoFOAM — a self-refining LLM agent that closes the semantic gap between natural-language intent and correct OpenFOAM simulation execution.

Key Contributions

- **Autonomous CFD Agent:**
Qwen2.5-Coder-14B fine-tuned on 252 CFD-specific prompts; syntactically sound and physically consistent output
- **Seven-Layer Evolution Loop:** SFT + DPO + anchor mixing + active learning + regression gate — monotonic performance improvement
- **Anti-Collapse Strategies:** RAG retry, surgical patching, prompt paraphrasing

Key Results

100% execution success
96.4% solver routing accuracy
13/13 geometry templates
7/7 solver families

Open Source

Code: github.com/AGN000/AutoFOAM
Data: github.com/AGN000/FoamAgentCases
Weights: [hf.co/arungovindneelan/foam-cfd-unified-14b](https://huggingface.co/arungovindneelan/foam-cfd-unified-14b)

Thank You

Questions & Discussion

Contact

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Resources

`github.com/AGN000/AutoFOAM`
`huggingface.co/arungovindneelan/...`